Linyi (Tiger) Hou

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Education

Ph.D. in Aerospace Engineering May 2025
University of Illinois Urbana-Champaign GPA: 3.94/4.00
Thesis: Autonomous Lost in Space and Time State Determination using Periodic Variable Stars

B.S. in Aerospace Engineering, Minor in Computer Engineering May 2021
University of Illinois Urbana-Champaign GPA: 3.93/4.00

Skills

Technical	batch filtering, Kalman filtering, Monte Carlo simulation, celestial navigation, orbit determination, spacecraft navigation modeling
Collaboration	technical/proposal writing, MS Office suite, LaTeX, Git, GitHub, Trello
Programming	Python, C/C++, MATLAB/Simulink, GMAT, Javascript, Julia

Research

Astrodynamics and Planetary Exploration Group (APEX) Advisor: Dr. Siegfried Eggl

Interstellar Navigation and Star Identification Sep 2024 – May 2025

- Developed a star identification technique using C++ which mitigates stellar aberration effects at relativistic speeds, improving star identification success rate from 38% to 91%.

Celestial Navigation to Supplement DSN Tracking Jul 2024 – Mar 2025

- Collaborated with colleagues to create navigation simulation tools in Python to assess the feasibility of reducing Deep Space Network tracking cadence by using onboard celestial navigation for the OSIRIS-APEX and CAPSTONE missions.
- Designed sensor scheduling algorithms for celestial navigation which reduced navigation uncertainty by over 10% compared to existing strategies in Monte Carlo simulation.
- Implemented spherical harmonics for an open source orbit propagator in C++.

Lost in Space and Time Navigation using Variable Stars Nov 2022 – Mar 2025

- Developed an autonomous optical navigation concept using variable star light curves which delivers timing accuracy within 1 second without needing prior position/time information.
- Simulated star tracker observations and demonstrated the feasibility of reestablishing Earth pointing via the proposed method through Monte Carlo analysis.

Aerospace Mission Analysis Laboratory Advisor: Dr. Zachary R. Putnam

Lost in Space X-ray Pulsar Navigation (XNAV) Jun 2021 – Dec 2023

- Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- Created a photon time of arrival simulation tool in Python and validated simulation results against the pulsar timing package PINT to within 50 µs.

Velocity-based Initial Orbit Determination — Aug 2019 – Apr 2024

- ❖ Developed a novel initial orbit determination technique using batched sequential range-rate measurements that achieved over 10x better accuracy than state-of-the-art methods.

Space Systems Optimization Laboratory — Advisor: Dr. Koki Ho

Architecture Trade Studies on ISRU Technologies for Human Space Exploration — Sep 2018 – Sep 2019

- ❖ Worked with two graduate students to discuss the development of a mixed-integer linear programming framework for space logistics optimization.
- ❖ Conducted literature review of in-situ resource utilization (ISRU) technologies to quantify input/output rates of technology options for an architecture optimization tool.

Work Experience

Teaching Assistant — Jun 2021 – May 2024

University of Illinois Urbana-Champaign (AE 202, AE 443, AE 483)

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Helped 20+ students formulate and implement UAV state-space model controllers with various sensors, e.g., laser rangefinders, optical flow cameras, and time-of-flight anchors.
- ❖ Assisted in developing LQR control-related code and instructional materials for a UAV course using C and Python.

Software Engineering Intern — May 2019 – Aug 2019

Collins Aerospace

- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated weekly builds and integration in TortoiseSVN using Python.

Projects

Collaborative Projects in Flight Dynamics — 2022, 2023

- ❖ Planetary entry: Created a 3-DOF entry simulation to assess the flight performance of several novel Apollo-derived entry guidance algorithms at Mars.
- ❖ Trajectory optimization: Applied optimal control theory to identify time- and fuel-optimal low-thrust Earth-Mars transfer trajectories.

Team Lead/Advisor, NASA's RASC-AL Competition — 2019, 2022, 2023

- ❖ Led a team of 10+ students to develop a reusable crewed lunar lander concept (structure, power, thermal, life support, orbital mechanics, budget, etc.).
- ❖ Advised two student teams who developed technical requirements and performed trade studies for lunar and Martian in-situ resource utilization architectures.

Honors and Awards

Aerospace Engineering Alumni Advisory Board Fellowship — 2024

Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46^{th} AAS GN&C Conference 2024
American Astronautical Society (AAS) Rocky Mountain Section

Yee Fellowship 2021, 2024
Grainger College of Engineering, University of Illinois Urbana-Champaign

David and Catherine Thompson Space Technology Scholarship 2020
American Institute of Aeronautics and Astronautics (AIAA)

H.S. Stillwell Problem Solving Award 2020
Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Robert W. McCloy Memorial Award 2019
Department of Aerospace Engineering, University of Illinois Urbana-Champaign

James Scholar 2018 - 2021
University of Illinois Urbana-Champaign

Dean's List 2017 - 2021
University of Illinois Urbana-Champaign

Journal Articles

[J.5] **L. Hou**, Ishaan Bansal, Clark Davis, and Siegfried Eggl. "Position and Time Determination without Prior State Knowledge via Onboard Optical Observations of Delta Scuti Variable Stars". In: *The Journal of the Astronautical Sciences* 72.9 (2025). doi: 10.1007/s40295-025-00485-8.

[J.4] **L. Hou** and Zachary R Putnam. "Autonomous Initial Orbit Determination using Sequential Range-Rate Measurements". In: *The Journal of the Astronautical Sciences* 71.24 (2024). doi: 10.1007/s40295-024-00442-x.

[J.3] **L. Hou** and Zachary R Putnam. "A Norm-Minimization Algorithm for Solving the Lost-in-Space Problem with XNAV". In: *The Journal of the Astronautical Sciences* 71.6 (2024). doi: 10.1007/s40295-023-00425-4.

[J.2] Hao Chen, Tristan Sarton du Jonchay, **L. Hou**, and Koki Ho. "Multifidelity Space Mission Planning and Infrastructure Design Framework for Space Resource Logistics". In: *Journal of Spacecraft and Rockets* 58.2 (2021), pp. 538–551. doi: 10.2514/1.A34666.

[J.1] Hao Chen, Tristan Sarton du Jonchay, **L. Hou**, and Koki Ho. "Integrated In-Situ Resource Utilization System Design and Logistics for Mars Exploration". In: *Acta Astronautica* 170 (2020), pp. 80–92. doi: 10.1016/j.actaastro.2020.01.031.

Conference Proceedings

[C.8] **L. Hou** and Siegfried Eggl. "Autonomous Optical Navigation using Astrometry and Photometry". In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 25-248. Kaua'i, HI, Jan. 2025.

[C.7] **L. Hou**, Ishaan Bansal, Clark Davis, and Siegfried Eggl. "Lost-in-space Position and Time Determination via Star Tracker Observations of Periodic Variable Stars". In: *46th Rocky Mountain AAS GN&C Conference*. AAS-24-012. Breckenridge, CO, Feb. 2024.

[C.6] Alec Auster, Ana Bojinov, Shikhar Kesarwani, Galen Sieck, Riya Shah, Madeline Odeen, Sara McCarthy, Madison Frankenthor, Komol Patel, Grant Davis, et al. "Mars Ice Thermal Harvesting Rig & ISRU Laboratory (MITHRIL)". In: *ASCEND 2022*. Aug. 2022, p. 4249. doi: 10.2514/6.2022-4249.

[C.5] **L. Hou** and Zachary R. Putnam. “A Norm-minimization Method for Solving the Cold-Start Problem with XNAV”. In: *AAS/AIAA Astrodynamics Specialist Conference*. AAS 22-560. Charlotte, NC, Aug. 2022.

[C.4] **L. Hou** and Zachary R. Putnam. “Initial Orbit Determination from Sequential Line-of-Sight Velocity Measurements”. In: *AAS/AIAA Astrodynamics Specialist Conference*. AAS 22-561. Charlotte, NC, Aug. 2022.

[C.3] **L. Hou**, Kevin Lohan, and Zachary R Putnam. “Comparison and Error Modeling of Velocity-Based Initial Orbit Determination Algorithms”. In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 21-280. Virtual, Feb. 2021.

[C.2] Hao Chen, Tristan Sarton du Jonchay, **L. Hou**, and Koki Ho. “Multi-Fidelity Space Mission Planning and Space Infrastructure Design Framework for Space Resource Logistics”. In: *AIAA Propulsion and Energy 2019 Forum*. 2019, p. 4134. DOI: 10.2514/6.2019-4134.

[C.1] H Chen, T Sarton du Jonchay, **L. Hou**, and K Ho. “Integrated Analysis Framework for Space Propellant Logistics: Production, Storage, and Transportation”. In: *Lunar ISRU 2019-Developing a New Space Economy Through Lunar Resources and Their Utilization* 2152 (2019), p. 5003. URL: https://www.hou.usra.edu/meetings/lunarisru2019/pdf/5003.pdf.